CSP 203: Software Tools and Technologies (Question 1)

Your Name (Roll Number)

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1 Introduction [6 Marks]

Live streaming is a powerful and direct way to get connected with thousands of concurrent viewers. No matter if you are using it for your product promotion, giving a motivating speech, or just want to stream the game.

1.1 Usage

It is a tool to circulate information in the form of videos. Users can enjoy the audio and video without waiting for their file to download.

1.2 Statistics

As live is on rise Halder et al. [1], the demand for high-quality and lower latency services is increasing. Thus, efficient, scalable, and economical solutions are required for live video streaming.

According to the report of Business Fortune Insights, the global video streaming market size is expected to grow from USD 342.44 billion in 2019 to USD 842.93 billion by 2027 at a compound annual growth rate of 12% Size [2].

2 Equation Section [9 Marks]

The score of a node is calculated after taking uploading bandwidth, video streaming rate, latency, and stability of that node into consideration.

1. Eqn. (1) has been proposed for the calculation of score.

$$Score(peer) = k_1*(\frac{upload_bandwidth}{streaming_rate})+k_2*(\frac{1}{latency})+k_3*(\frac{active_duration}{num_of_failure}) \quad (1)$$

2. The general Tailor Series expansion is given in the form (2):

$$f(x) = f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f^{(3)}(a)}{3!}(x-a)^3+\cdots+\frac{f^{(n)}(a)}{n!}(x-a)^n+\cdots \quad (2)$$

3. The above expansion can be expressed as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

3 Table [5 Marks]

Table 1: Performance Comparison of Algorithms

Algorithm	**Time Complexity**	**Space Complexity**
Linear Search	$O(n)$	$O(1)$
Binary Search	$O(\log n)$	$O(1)$
Merge Sort	$O(n \log n)$	$O(n)$
Quick Sort	$O(n \log n)$	$O(\log n)$

4 Algorithm [5 Marks]

The following algorithm-(1) is used to search an element in a sorted array in $O(log(n))$ time.

```
low ← 0;
high ← n − 1;
while low ≤ high do
    mid ← (low + high)/2;
    if A[mid] == key then
        return mid;
    else
        if A[mid] < key then
            low ← mid + 1;
        end
        else
            high ← mid − 1;
        end
    end
end
return −1;
```

Algorithm 1: Binary Search Algorithm

5 Image [5 Marks]



Figure 1: Linux Desktop Image

The above fig.1 can be seen after we log in to the ubuntu operating system.

6 Bibliography [5 Marks]

References

[1] Debajyoti Halder, Prashant Kumar, Saksham Bhushan, and Anand M Baswade. Fybrrstream: A webrtc based efficient and scalable p2p live streaming platform. In *2021 International Conference on Computer Communications and Networks (ICCCN)*, pages 1–9. IEEE, 2021.

[2] Graphene Market Size. Video streaming market size. *Share & Trends Analysis Report by Streaming Type, by Solution, by Platform, by Service, by Revenue Model, by Deployment Type, by User and Segment Forecasts*, 2027, 2020.